

D335

Introduction to Programming in Python

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A student writing a script for a pizza shop that opened in 2015 needs to split 17 slices among 5 people. She writes `whole = 17 // 5` and `left = 17 % 5`. The shop's owner wants the script finished by Friday. What are the values of `whole` and `left`?

- A. `whole = 2`, `left = 3`
- B. `whole = 3`, `left = 2`
- C. `whole = 3`, `left = 0`
- D. `whole = 3.4`, `left = 2`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A variable `n` holds a whole number. Which expression is True exactly when `n` is even?

- A. `n // 2 == 0`
- B. `n / 2 == 0`
- C. `n % 2 == 1`
- D. `n % 2 == 0`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A clinic scheduling script, written by a volunteer who also coaches soccer, prints appointment slots with `for slot in range(2, 8): print(slot)`. The clinic sees about 40 patients a day. How many numbers are printed, and what is the last one?

- A. 8 numbers; the last is 7.
- B. 7 numbers; the last is 8.
- C. 6 numbers; the last is 7.
- D. 6 numbers; the last is 8.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A list of warehouse bins is `bins = ['A1', 'B2', 'C3', 'D4', 'E5']`. The warehouse is 60,000 square feet. What does `bins[1:3]` return?

- A. `['A1', 'B2', 'C3']`
- B. `['A1', 'B2']`
- C. `['B2', 'C3']`
- D. `['B2', 'C3', 'D4']`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A list `scores` holds 10 values. Which expression returns the last value?

- A. `scores[-1]`
- B. `scores[0]`
- C. `scores[10]`
- D. `scores[1]`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A developer who just switched from spreadsheets writes: `def total(a, b): print(a + b)`. Then she runs `result = total(4, 6)`. Her manager asked for this in the quarterly report. What is stored in `result`?

- A. 10
- B. An error, because functions must always return a value.
- C. `'46'`
- D. None

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A function should give its caller the area of a rectangle so the value can be used in a later calculation. Which body does that?

- A. `print(width * height)`
- B. `area = width * height`
- C. `return width + height`
- D. `return width * height`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A program stores a customer code as `code = 'abc123'` and tries `code[0] = 'X'`. The code format was chosen by a vendor in 2019. What happens?

- A. A `TypeError`, because strings cannot be changed in place.
- B. An error, because the vendor's format is locked.
- C. code becomes 'X'.
- D. code becomes 'Xbc123'.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A script for a library that serves three towns has `count = 5` at the top. A function `def reset(): count = 0` is defined and then called with `reset()`. The library closes at 8 p.m. What does `print(count)` show after the call?

- A. None
- B. 5
- C. An error, because `count` is defined twice.
- D. 0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A loop checks order totals `[20, 45, 0, 80, 15]`. Inside, it runs `if t == 0: break` and otherwise adds `t` to a running sum that starts at 0. The store runs a sale every March. What is the sum when the loop ends?

- A. 160
- B. 65
- C. 20
- D. 0, because `break` throws away the running sum.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A loop over [3, -1, 4, -2, 5] should add up only the positive numbers and keep going to the end. Which statement, placed under `if n < 0:`, does that?

- A.** `return`
- B.** `break`
- C.** `continue`
- D.** `pass`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A data analyst at a fast-growing startup writes `import math` at the top of a script, then calls `sqrt(49)`. The startup has 30 employees. What happens?

- A.** A `NameError`, because `sqrt` must be written `math.sqrt`.
- B.** It returns 7.
- C.** It returns 7.0.
- D.** It installs the `math` package from the internet.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A student writing a script for a pizza shop ~~that opened in 2015~~ needs to split 17 slices among 5 people. She writes `whole = 17 // 5` and `left = 17 % 5`. ~~The shop's owner wants the script finished by Friday.~~ What are the values of `whole` and `left`?

Target word: values of `//` and `%` **Decides:** `17 // 5` is the floor quotient; `17 % 5` is the remainder.

A	Inverted operation	Swaps the quotient and remainder.
B	Correct	<code>17 = 5 × 3 + 2</code> .
C	Dropped element	Ignores the 2 slices left over.
D	Confusion set	3.4 is <code>17 / 5</code> ; <code>//</code> drops the fraction.

2. Answer D Concept 1 · transfer

A variable `n` holds a whole number. Which expression is `True` exactly when `n` is even?

Target word: choose the operator for evenness **Decides:** Even means the remainder after dividing by 2 is 0.

A	Confusion set	Floor division is 0 only for <code>n = 0</code> or 1, not for all even numbers.
B	Adjacent category	<code>True</code> only when <code>n</code> is 0.
C	Inverted operation	That tests for odd.
D	Correct	<code>%</code> gives the remainder; even numbers leave 0.

3. Answer C Concept 2

A clinic scheduling script, ~~written by a volunteer who also coaches soccer~~, prints appointment slots with `for slot in range(2, 8): print(slot)`. ~~The clinic sees about 40 patients a day.~~ How many numbers are printed, and what is the last one?

Target word: count and last value of range **Decides:** range excludes the stop value 8.

A	Wrong stem number	Uses the stop value as the count.
B	Absolute language	Treats the stop value as included.
C	Correct	2, 3, 4, 5, 6, 7 — six values, ending before 8.
D	Dropped element	Right count, but 8 itself is never produced.

4. Answer C Concept 3

A list of warehouse bins is `bins = ['A1', 'B2', 'C3', 'D4', 'E5']`. ~~The warehouse is 60,000 square feet.~~ What does `bins[1:3]` return?

Target word: slice result **Decides:** Index 1 is the second item; the slice stops before index 3.

A	Inverted operation	Treats indexing as starting at 1 and includes the end.
B	Confusion set	Counts the first element as index 1.
C	Correct	Indexes 1 and 2; 3 is excluded.
D	Absolute language	Includes index 3, the excluded end.

5. Answer A Concept 3 · transfer

A list scores holds 10 values. Which expression returns the last value?

Target word: index of the last element **Decides:** Indexes run 0 to 9; -1 counts from the end.

A	Correct	-1 is always the last element, and scores[9] would also work.
B	Inverted operation	Index 0 is the first value.
C	Wrong stem number	Uses the length as an index; the last index is 9, so this raises IndexError.
D	Confusion set	Index 1 is the second value.

6. Answer D Concept 4

A developer ~~who just switched from spreadsheets~~ writes: `def total(a, b): print(a + b)`. Then she runs `result = total(4, 6)`. ~~Her manager asked for this in the quarterly report.~~ What is stored in result?

Target word: value stored from a print-only function **Decides:** The function has no return statement.

A	Confusion set	10 is printed, not returned.
B	Absolute language	Functions without return are legal; they return None.
C	Adjacent category	String joining happens only with strings, and nothing is returned anyway.
D	Correct	Printing shows 10 on screen, but without return the function gives back None.

7. Answer D Concept 4 · transfer

A function should give its caller the area of a rectangle so the value can be used in a later calculation. Which body does that?

Target word: choose return over print **Decides:** The caller needs the value, not screen output.

A	Confusion set	Shows the value but gives the caller None.
B	Dropped element	Computes a local value but never sends it back.
C	Inverted operation	Adds instead of multiplying.
D	Correct	return sends the value back for later use.

8. Answer A Concept 5

A program stores a customer code as `code = 'abc123'` and tries `code[0] = 'X'`. ~~The code format was chosen by a vendor in 2019.~~ What happens?

Target word: result of item assignment on a string **Decides:** Strings are immutable.

A	Correct	Build a new string instead, e.g., <code>'X' + code[1:]</code> .
B	Camouflage	The vendor detail has nothing to do with Python's type rules.
C	Dropped element	Ignores that this is item assignment, not reassignment of the whole variable.
D	Adjacent category	That would work for a list, not a string.

9. Answer B Concept 6

A script for a library that serves three towns has `count = 5` at the top. A function `def reset(): count = 0` is defined and then called with `reset()`. ~~The library closes at 8 p.m.~~ What does `print(count)` show after the call?

Target word: effect of a local assignment on an outer variable **Decides:** `count = 0` inside the function creates a new local variable.

A	Confusion set	None is what <code>reset()</code> returns, not what <code>count</code> holds.
B	Correct	The function's <code>count</code> is local; the outer <code>count</code> is unchanged.
C	Absolute language	Using the same name in two scopes is allowed.
D	Inverted operation	Would require a global declaration.

10. Answer B Concept 7

A loop checks order totals `[20, 45, 0, 80, 15]`. Inside, it runs if `t == 0`: `break` and otherwise adds `t` to a running sum that starts at 0. ~~The store runs a sale every March.~~ What is the sum when the loop ends?

Target word: sum after `break` **Decides:** `break` exits at the 0, so only 20 and 45 are added.

A	Confusion set	Treats <code>break</code> like <code>continue</code> and adds 80 and 15.
B	Correct	$20 + 45 = 65$, then <code>break</code> stops the loop.
C	Dropped element	Stops one value too early.
D	Inverted operation	<code>break</code> stops the loop; it does not undo work already done.

11. Answer C Concept 7 · transfer

A loop over `[3, -1, 4, -2, 5]` should add up only the positive numbers and keep going to the end. Which statement, placed under `if n < 0`:, does that?

Target word: choose the skip statement **Decides:** Skip the negatives but do not exit.

A	Adjacent category	Leaves the whole function, not just one pass.
B	Inverted operation	Exits at <code>-1</code> and gives 3.
C	Correct	Skips to the next number; the sum becomes 12.
D	Confusion set	Does nothing, so the negative would still be added if the <code>add</code> follows.

12. Answer A Concept 8

A data analyst at a fast-growing startup writes `import math` at the top of a script, then calls `sqrt(49)`. ~~The startup has 30 employees.~~ What happens?

Target word: name access after `import` **Decides:** `import math` requires the `math.` prefix.

A	Correct	<code>import math</code> loads the module name, not its functions directly.
B	Adjacent category	Even with the prefix, <code>math.sqrt</code> returns a float.
C	Confusion set	That would work only with <code>from math import sqrt</code> .
D	Inverted operation	<code>import</code> loads an installed module; installing is done with <code>pip</code> .

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Division operators	/ always returns a float; // returns the floor quotient; % returns the remainder.	Expecting 7 / 2 to give 3, or mixing up // and %.	An expression with /, //, or % between two integers.
2	range() stops before the end	range(start, stop) includes start and excludes stop; range(n) runs n times from 0 to n-1.	Counting the stop value as included.	A loop over range() and a question about the last value or count.
3	Zero-based indexing and slicing	The first element is index 0; a slice s[a:b] includes a and excludes b; -1 is the last element.	Treating index 1 as the first element or including the slice end.	Square brackets with numbers after a list or string.
4	return vs. print	return sends a value back to the caller; a function with no return gives None, even if it prints.	Thinking a printed value is stored in the variable.	x = func() where func prints but does not return.
5	Mutable vs. immutable types	Lists and dictionaries can change in place; strings, tuples, and numbers cannot.	Trying to assign to one character of a string.	An item assignment like s[0] = ... on a string or tuple.
6	Variable scope	A variable assigned inside a function is local to it unless declared global; it disappears when the function ends.	Expecting a function's local assignment to change the outer variable.	The same name assigned both inside and outside a function.
7	Loop control	break exits the loop immediately; continue skips to the next pass; a while loop needs its condition to become False to stop.	Swapping break and continue.	break or continue inside an if within a loop.
8	Importing modules	import m requires m.name to use contents; from m import name brings name in directly.	Calling sqrt() after only import math.	An import line followed by a function call with or without the module prefix.

Course scope (public description): Introduction to Programming in Python introduces skills in creating Python scripts with basic programming concepts. Learners will be able to create control flow with functions and loops, and to implement code with packages, modules, and libraries.